MAHARSHI OZA

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PROFESSIONAL SUMMARY

Verification engineer with industry RTL simulation experience in SystemVerilog/UVM and hands-on system validation of automotive ECU hardware. Builds Python and Makefile automation for large simulation and implementation flows, and works across RTL design, synthesis, timing and hardware bring-up. MS in Computer Engineering, NC State.

CORE COMPETENCIES

Verification & Simulation: RTL Simulation, UVM, SystemVerilog, SVA, Constrained-Random Testing, Functional & Code Coverage, Scoreboards, Regression & Debug, Gate-Level Simulation
Automation & Scripting: Python (regression, report parsing, data analysis), Makefile, Tcl, Perl, Shell; automated multi-run tool flows and result collection
System Validation: ECU bench bring-up, JTAG and GDB debug, CAN-bus analysis (Vector CANoe/CANalyzer), requirements-based KPI validation, recorded-data testing
Design & Timing: Verilog/SystemVerilog RTL, AXI4, I2C, Synthesis, Static Timing Analysis, CDC, Floorplan/Place/CTS/Route, DFT, Low Power (UPF)
Tools: VCS, Xcelium, QuestaSim, Icarus Verilog, Design Compiler, ICC2, Genus, Vivado, Git, Linux

PROFESSIONAL EXPERIENCE

RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India **Aug 2023 – Aug 2024**

- Implemented an out-of-order scoreboard for an AXI4 memory subsystem in SystemVerilog/UVM, tracking outstanding transactions by ID and checking read/write data, responses and per-ID ordering.
- Developed functional coverage for AXI4 bursts, IDs, out-of-order completion and handshake stalls (over 90%); verified reset, FIFO full/empty and handshake corner cases and debugged regressions in Questa, Xcelium and VCS.
- Maintained UVM environments for Asynchronous FIFO and FPU blocks and contributed to multi-port packet router RTL (FSM control, FIFOs, synchronizers).

Functions Engineer – Vision & Perception | Magna Electronics, India **Dec 2021 – Aug 2023**

- Implemented requirements-based KPI checkers in C++ inside the ADAS Lane Marker Detection application, translating SRD/SRS acceptance thresholds into self-checking pass/fail logic for lane type, color, lateral distance to the rear axle and curvature against ground truth.
- Functionally validated the classical image-processing and DNN (ONNX) lane-detection pipelines on recorded camera and CAN datasets, reusing the same KPI checks to compare both implementations; supported data annotation for the DNN transition.
- Supported ECU bench bring-up with a senior engineer: deployed the C++ binary to target hardware and debugged input-data issues with GDB breakpoints and a JTAG debugger; analyzed vehicle CAN signals in Vector CANoe/CANalyzer.
- Implemented code, branch, and line coverage for perception modules and the KPI framework to expose validation gaps; all development followed MISRA C/C++ compliance.

PROJECTS

UVM Verification of a Set-Associative Write-Back Cache Controller | SystemVerilog, UVM, SVA, Python

- Built a UVM 1.2 environment for a parameterized write-back/write-allocate cache controller with 3 agents (CPU, flush, reactive memory responder via TLM analysis FIFO), a reference-model scoreboard, and SVA protocol checks split into testbench-side and DUT-side assertions.
- Developed 7 directed and constrained-random tests using factory config overrides (backpressure, flush/CPU races, memory error injection); built a Python regression flow merging per-run coverage and reporting holes, with 0 data mismatches across 1,000+ accesses under injected errors.

Physical Design Space Exploration – ISCAS-89 s38584 | Design Compiler, ICC2, GF180MCU, Python, Makefile, Tcl

- Took the s38584 benchmark (~1,250 flops, ~7K cells) through synthesis with scan configuration, floorplan, PG mesh, placement, CTS and routing; automated 85 parallel runs with Makefiles and Python, parsing 45 metrics per run with custom STA and ADL checks.
- Final configuration (5 ns, 0.8 utilization, Metal3) closed timing with 0 DRCs and positive WHS at ~256.8K μm^2 and ~94% utilization, cutting the area × delay × layer metric from 17.7M to 3.1M (~5.7×).

Functional Verification of LC-3 Processor | UVM/UVMF

- Architected a reusable UVM/UVMF environment covering LC-3 pipeline stages and instruction/data memory; developed sequences, tests, assertions, regression, and coverage models for control-flow, datapath, and corner cases, achieving 100% functional coverage closure.

UVM-Based Verification of AXI4 Slave | SystemVerilog

- Built a UVM environment for a 32-bit AXI4 slave supporting FIXED/INCR bursts; implemented agents, scoreboard, SVA protocol checks, and functional coverage for burst types, sizes, and boundary conditions, including constrained-random backpressure scenarios.

Scan Insertion & ATPG Setup: 8-bit Counter (Cadence DFT Fundamentals labs) | Genus, Modus, Verilog, Icarus

- Inserted mux-D scan into an 8-bit counter in Cadence Genus (DFT DRC, one auto-stitched chain) and generated the scan netlist, SCANDEF and Modus ATPG scripts.
- Verified scan shift/capture on the gate-level netlist with a self-checking Verilog testbench (load → capture → unload).

Pipelined Transformer Self-Attention Accelerator | SystemVerilog, Design Compiler

- Designed an SRAM-based self-attention accelerator (Q/K/V projections, score and attention matrices); verified against reference results with 217/217 checks passing across 4 test cases and closed timing at 5 ns in Nangate 45 nm.

EDUCATION

M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4 **Aug 2024 – May 2026**
Relevant Coursework: Advanced Verification using UVM, ASIC Verification, ASIC and FPGA Design with Verilog, Physical Design, Advanced Microprocessor Architecture

B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India **Aug 2018 – May 2022**